

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL2_ Scenario Based Assignment

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Code of Conduct:

*I Ganji Madhan Kumar (192472328) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

G.Madhan Kumar

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario:** A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments
- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario

- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Understanding of Scenario	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	10
Identification of Key Issues	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	10
Application of Concepts	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	12.5
Decision Making	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	10

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
Clarity of Response	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	7.5
				Total	50

1. Behavior of Greedy Best-First Search:

Greedy Best-First Search (GBFS) selects the next node based only on the heuristic value $h(n)$. It always chooses the location that appears closest to the destination without considering the cost already spent.

Working in this Scenario

- The drone starts from the medical center.
- It estimates the straight-line distance to the hospital.
- At every step, it chooses the path with the smallest estimated distance.
- It ignores fuel usage, weather conditions, flooded roads, and previous travel cost.

Example

Suppose there are two possible routes.

Route	Straight-line Distance	Actual Cost
Route A	8 km	Heavy rain, flooded roads
Route B	10 km	Clear weather

GBFS selects Route A because it appears closer, even though it is actually slower and riskier.

Strengths

- Very fast decision making.
- Low computational overhead.
- Suitable when immediate action is required.
- Requires less memory compared to A*.

Weaknesses

- Does not guarantee the shortest or safest path.
- Easily misled by inaccurate heuristic estimates.
- May repeatedly move toward blocked routes.
- Performs poorly in highly dynamic environments.

2.Behavior of A* Search

A* Search considers both:

- $g(n)$: Actual cost from the start
- $h(n)$: Estimated cost to the goal

Its evaluation function is

$$f(n) = g(n) + h(n)$$

Instead of choosing only the nearest location, A* evaluates the total estimated cost.

Working in this Scenario

Suppose the drone has two choices.

Route	$g(n)$	$h(n)$	$f(n)$
Route A	12	8	20
Route B	9	12	21

Although Route A has a lower heuristic value, A* also considers the travel cost already incurred before making the decision.

If a flooded road increases the travel cost, A* automatically updates the route evaluation and searches for a better alternative.

Advantages

- Finds the optimal path when the heuristic is admissible.
- Considers both travel cost and estimated remaining distance.
- Produces safer and more reliable routes.
- Adapts better to changing movement costs.

Disadvantages

- Uses more memory.
- Computationally more expensive than GBFS.
- May require replanning when the environment changes rapidly.

3.Impact of Heuristic Accuracy

The heuristic function estimates how far the goal is from the current position.

Accurate Heuristic

If the estimate closely matches the actual distance:

- Greedy Search quickly reaches the destination.
- A* expands fewer nodes.
- Faster route planning.
- Lower computational cost.

Inaccurate Heuristic

If the estimate is poor:

- Greedy Search may choose dangerous or blocked paths.
- A* still considers actual travel cost, making it more reliable.
- More nodes may need to be explored.
- Search time increases.

Comparison

Heuristic Accuracy	Greedy Search	Best-First	A* Search
Accurate	Very Fast		Fast and Optimal
Slightly Inaccurate	May choose wrong path		Usually still optimal
Highly Inaccurate	Performs poorly		Slower but safer

4. Effect of Dynamic Changes (Blocked Paths)

Flood conditions change continuously.

Examples include:

- Bridges collapsing
- Roads becoming flooded
- Strong winds increasing travel time
- Restricted flying zones

Effect on Greedy Best-First Search

- Continues following the heuristic.
- May repeatedly approach blocked paths.
- Requires frequent replanning.
- Often wastes time.

Effect on A*

- Updates movement costs.
- Recalculates the optimal path.
- Avoids expensive or blocked routes.
- Produces safer navigation.

Comparison

Dynamic Situation	Greedy	A*
New obstacle appears	Often fails or backtracks	Replans efficiently
Weather worsens	Ignores actual cost	Updates path cost
Variable terrain	Less effective	More reliable

5. Recommended Algorithm

Recommended Algorithm: A* Search

Justification

Although Greedy Best-First Search makes faster decisions, it ignores actual travel costs and may select risky or blocked routes.

A* Search is more suitable because it:

- Balances actual travel cost and estimated distance.
- Finds the optimal path using an admissible heuristic.

- Handles changing weather and terrain more effectively.
- Reduces delivery delay caused by blocked routes.
- Improves mission safety during emergency operations.